

Precise Opening Detection of Heart Aortic Valve Towards Noninvasive Continuous Blood Pressure Detection

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Abstract—This paper presents a novel earbud-based system for continuous, noninvasive blood pressure (BP) monitoring using precise arterial pulse transit time (PTT) estimation. The system integrates a low-noise inertial measurement unit (IMU) and a photoplethysmography (PPG) sensor, achieving highly synchronized data acquisition at 450 Hz with <1.1 millisecond synchronization error between the sensors. Key contributions include: (i) a rigorous validation of ballistocardiogram (BCG)-based proximal timing (B-point) estimation against ground truth B-point derived from synchronized electrocardiography (ECG) and impedance cardiography (ICG); (ii) development of a custom earbud prototype optimized for BP monitoring with high-precision arterial PTT measurement; and (iii) introduction of a B-point estimation algorithm leveraging multi-axis BCG fusion to enhance signal-to-noise ratio (SNR) and improve accuracy. Results show that BCG fusion reduces mean absolute error (MAE) in B-point estimation by 5.33% and 6.70% compared to using only the X- or Y-axis BCG signals, respectively. Furthermore, using the I-wave of the BCG signal as a proxy for the opening of the aortic valve reduces the MAE by 59.45% and 32.09% compared to using the H-wave and the J-wave, respectively. These findings establish a foundation for practical, wearable, and noninvasive continuous BP monitoring, with potential applications in real-world cardiovascular health management.

I. INTRODUCTION

Cardiovascular disease remains one of the leading causes of morbidity and mortality worldwide [1]. Continuous monitoring of blood pressure (BP) is essential for early detection, effective management, and prevention of complications related to hypertension [2]. However, conventional cuff-based devices are impractical for long-term continuous monitoring due to their intermittent operation, discomfort, and lack of real-time data. To address these limitations, recent research has focused on non-invasive wearable BP monitoring systems based on pulse transit time (PTT) measurements [3]–[7].

PTT is defined as the time it takes an arterial pulse wave to travel between two measurement sites in the vascular system. It is highly sensitive to changes in arterial stiffness and BP, making it a widely pursued biomarker for cuffless BP monitoring [8], [9]. However, key challenges remain to reliably capture the mechanical events of the heart and accurately measure both pulse initiation (proximal pulse timing) and pulse arrival (distal pulse timing) using a single wearable device. First, BP estimation based on PTT is very sensitive to the timestamp mismatch between the proximal and distal

time markers. Even if the signals are collected from the same device, there is no guarantee that the different sensors share the synchronized clock. Second, although studies attempt to link ballistocardiogram (BCG) morphology to the opening of the aortic valve from a physiological interpretation perspective [10], [11], using extensive experimental data for rigorous comparisons between BCG signals and the ground truth opening time of the aortic valve remain limited. Addressing this limitation is likely to provide significant opportunities for enhancing the accuracy and reliability of PTT-based BP estimation.

To address these limitations, this paper presents an earbud-based system, integrating a highly synchronized ultra-low-noise inertial measurement unit (IMU) and a photoplethysmography (PPG) sensor. This design enables the precise detection of the opening of the aortic valve (proximal pulse timing) and the arrival of the pulse in the concha region of the ear (distal pulse timing). Furthermore, we introduce a multimodal proximal timing estimation algorithm, which significantly improves the accuracy of detecting the true mechanical onset of blood ejection from the heart.



Fig. 1. The wearable sensor locations on the prototype of the earbuds.

In this study, we conduct the first comprehensive comparison between different fiducial points on the BCG waveform and the ground truth on the opening time of the aortic valve (B-point), which is derived from synchronized Electrocardiography (ECG) and Impedance Cardiography (ICG). Our

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analysis confirms that the I-wave of the BCG signal is the most reliable proxy for aortic valve opening. In addition, BCG fusion from different axes can further enhance the signal-to-noise ratio (SNR), based on which a more reliable B-point proxy can be derived.

The key contributions of this work are as follows.

- **Rigorous validation of BCG-based B-point estimation using ear-worn sensors:** We systematically evaluate fiducial points within the ear-based BCG waveform against validated ground-truth measurements of aortic valve opening, demonstrating the I-wave as a reliable proximal timing marker for earbud-based PTT and BP estimation.
- **Development of a custom earbuds prototype optimized for BP monitoring:** We design and implement an earbuds-based wearable system with a high-sensitivity IMU and synchronized PPG sensor, adapted to detect the opening of the aortic valve and the arrival of pulses in a single compact device.
- **Multi-axis BCG fusion algorithm for enhanced B-point timing detection:** We introduce a BCG fusion approach that integrates signals from multiple IMU axes, improving the signal quality and robustness of aortic valve opening detection in an ear-worn sensing system.

By addressing these challenges, this work establishes a foundation for non-invasive and continuous BP monitoring using a single wearable device.

II. METHODS

A. Earbuds Prototype Design

Unique requirements must be met for arterial PTT to be captured with high accuracy. This emphasizes the necessity of submillisecond precision. Two important requirements must be addressed. 1) a sufficiently high sampling rate to ensure quantization errors are below 2 ms, and 2) highly synchronized sensing between PPG and BCG signals, maintaining synchronization error below 2 ms. Additionally, as the mechanical vibration induced by the aortic valve opening can be very subtle, we need an accelerometer with ultra-low noise floor. Without considering these factors, the extracted PTT will be largely contaminated by sampling errors. A robust methodology is also needed to precisely synchronize the wearable sensing signals with external reference signals (including ECG and ICG) to support accurate aortic valve opening detection.

There are no existing wearable devices on the market that meet all these requirements. The PPG and IMU sensors in the commercialized device are usually integrated with their respective front-end and thus are sampled at different paces. The data points from the different sensors are packeted and assigned to the same timestamps based on their arrival at the Microcontroller Unit (MCU) or Bluetooth Low Energy (BLE) chipset, although their underlying sampling instances are different. Meanwhile, considering the battery life and lighter demands for functions such as heart rate detection,

sampling rates less than 50 Hz are usually used. The main purpose of the IMU sensor in commercialized devices is to track user's motion. Therefore, they usually focus on a large detection range instead of a low noise floor.

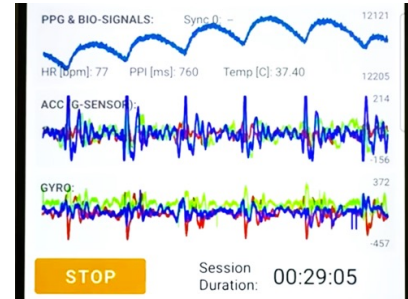


Fig. 2. The app showing displaying PPG, 3-axis accelerometer, and 3-axis gyroscope signals while a participant was sitting stationary in a chair. The blood pulse and the J-wave on the accelerometer's Y-axis are clearly visible.

We built the earbuds dedicated to address these issues. The device is shown in Fig. 1 and the key specifications are listed in Table I.

The MCU (SBS3000 WLCSP) simultaneously triggers the PPG (IR LED: SFH4043; Green: 16-213/GHC-YR1S1/3T; 2 × Photodetectors: Nisshinbo NJL6193; analog front end: TI AFE44I30) and IMU (ST LSM6DSO, includes a 3-axis accelerometer and 3-axis gyroscope) sensors at a variable sampling rate, up to 450 Hz. To maximize the sensing capability of the accelerometer in capturing subtle mechanical vibrations caused by the opening of the aortic valve, we configured its noise floor to $70 \mu\text{G}/\sqrt{\text{Hz}}$, with 16-bit resolution. An out-of-the-shelf true wireless stereo (TWS) enclosure is modified to house the devices. Importantly, the PCB is mounted so that the accelerometer's Y-axis is aligned with head-foot directions so that maximum heartbeat-induced vibration can be captured. We designed a dedicated general-purpose input/output (GPIO) port with a mini-USB connector to receive stimuli in the form of TTL pulses from external equipment. The time marker associated with the stimuli can then serve as a flag to align the signals recorded by the earbuds and external equipment.

The data are continuously streamed to an ultra-low power bluetooth wearable platform (BES2700L), embedded with a BLE transceiver. The sensing data are sent to a paired Samsung Galaxy S24+ mobile phone. The raw data are saved in the memory as comma separated value (csv) files. In order to enable the study coordinators to monitor the signal quality, and to adjust the participants' posture in real time, we visualize the PPG and BCG signals on the app user interface (UI). We filtered the BCG signal using an one-order two-way Butterworth filter with cut off frequency [0.5, 10] Hz so that the significant fiducial points are visible on the app for the coordinators. An example of the app UI is shown in Fig. 2. The J-peaks on the accelerometer's Y-axis, which are slightly time-leading the PPG diastolic valleys are clearly visible.

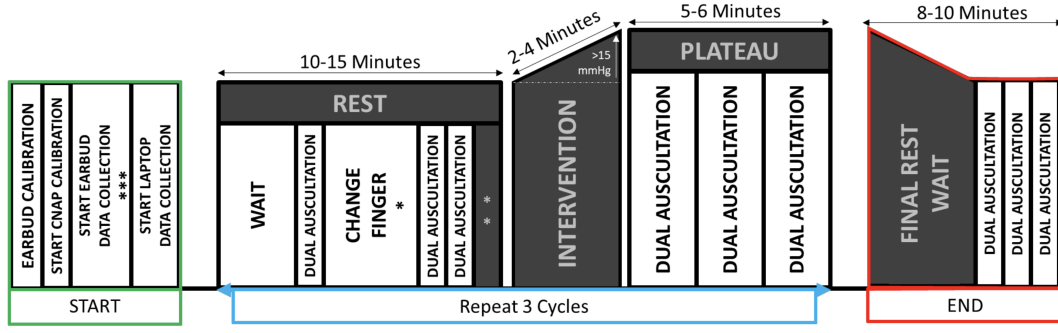


Fig. 3. Study protocol illustration. It includes three cycles of Rest-Intervention-Plateau stages: two cycles of leg-pressing exercises and one cycle of cold pressor test.

TABLE I

SUMMARY OF THE KEY SPECIFICATIONS OF THE EARBUDS PROTOTYPE

Sensor	Parameters	Prototype Performance
BCG	Sampling Rate Precision	Tunable, max 450 Hz Tunable, min 70 $\mu\text{G}/\sqrt{\text{Hz}}$
Sync betw. BCG-PPG	Sync error	1 millisecond
PPG	Sampling rate	Tunable, max 450 Hz
Battery	Charge interval	24 h PPG 450 Hz cont. on
External Trigger Flag	Sync error	< 1 millisecond

B. Data Collection

We designed a clinical study with the aim of collecting synchronized data from the earbuds (including PPG and BCG), the reference device (including ECG and ICG) and the ground truth measurement (including auscultation and non-invasive continuous blood pressure) from healthy subjects, while inducing a large variation in blood pressure. Seventy subjects were recruited representing a wide range of ages and activity levels. The study was approved by the Institutional Review Board (Salus IRB, protocol number: PR2024-600).

The experimental paradigm includes three cycles of Rest-Intervention-Plateau stages as illustrated in Fig. 3. We adopted two types of tasks to induce elevated blood pressure, based on two different physiological mechanisms. In the first two cycles, the subjects performed leg-pressing exercises, which can induce elevation of BP through increased cardiac output [12]. In the last cycle, participants immersed their feet in ice water with a closely monitored temperature of 5-10 centigrade, which can induce elevation of BP through increased total peripheral resistance [13].

Along with the wearable PPG and IMU signals, the reference ECG and ICG signals are collected using BIOPAC data acquisition system (BIOPAC System Inc., Goleta, CA, USA), a research grade instrument, with a sampling rate of 1250 samples / second [14]. The blood pressure ground truth is collected using two different measures. First, as the U.S. Food and Drug Administration (FDA) accepted gold standard, three auscultations are measured in each of the three stages. Each of the measurements are observed independently by two auscultators, from which the average of

the two measurements is recorded. Second, we used a finger clamp-based instrument, CNAP (CNSystems Medizintechnik GmbH., Graz, Austria), to record continuous BP values such that a large set of data points can be obtained to train a generalizable BP prediction model. A specially designed TTL stimuli train is configured in the BIOPAC system, from which we can align the signals from different devices at the millisecond level.

III. RESULTS

A. Synchronization between BCG and PPG on Earbuds

We first validate that the IMU and PPG signals collected from the earbuds are correctly time-synchronized. We acquired the PPG and IMU signals simultaneously when placing the earbuds on a stationary platform. We introduce a sudden and sharp impact on the platform. The data were acquired at a sampling interval of 2.2 ms (1/450 samples/second). As visualized in Fig. 4, the PPG and IMU signals (only the accelerometer Y-axis is visualized) start jumping at the same time index, which implies that the synchronization error is smaller than half of the sampling interval, i.e., 1.1 ms.

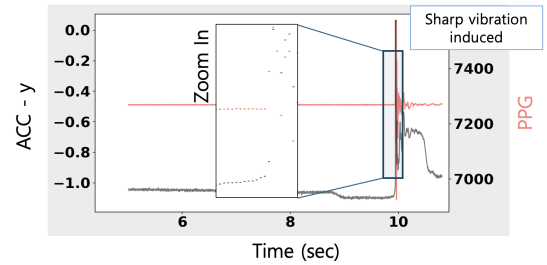


Fig. 4. Synchronization between PPG and IMU signals is smaller than 1.1 ms.

B. Estimation of B-point from BCG and PPG on Earbuds

We used the approach proposed by Lozano et al. [15] to establish the ground truth time marker for the opening of the aortic valve (B-point), jointly using the synchronized ECG and ICG signals. ECG provides the R-wave as a temporal reference, while the ICG provides the dZ/dt signal, which is the derivative of the impedance signal. First, the ECG

and dZ/dt signals are band-pass filtered using a first-order forward-backward Butterworth filter, with cutoff at [0.6, 40] Hz and [0.5, 15] Hz, respectively. The time between the R-wave and the B-point is reliably estimated using a polynomial function

$$RB = 1.233RZ - 0.0032RZ^2 - 31.59, \quad (1)$$

where RB and RZ are the time distances between R wave (from filtered ECG) and B-point, and between R wave and C point (prominent peak on dZ/dt signal), respectively. This approach leverages the physiological relationship where the same ventricular pressure function influences both valve opening and peak ventricular ejection, and is more robust than methods relying on second or third derivatives of dZ/dt . It is validated across various conditions and populations, serving as a robust reference for evaluating other techniques [15]. An example of the determined B-points and the associated R wave of the ECG signal, the C peak of the dZ/dt signal, are illustrated in Fig. 5.

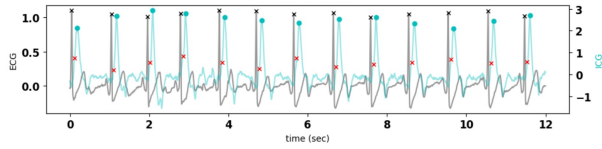


Fig. 5. Establishing the ground truth marker for the opening of the aortic valve (B-point). The cyan circles represent the C-peaks of the dZ/dt signal. The black x marks represent the R waves of the ECG signal. The red x marks represent the ground truth B-points.

C. Signal Processing and B-point Estimation from Earbuds

We develop an algorithm for the detection of earbuds B-points based on synchronized BCG and PPG signals. After visual inspection along with PPG, we decided to focus on the X-axis and Y-axis accelerometer signals because they present visible time markers associated with blood pulses. First, the PPG and BCG signals are bandpass filtered using the first-order forward-backward Butterworth filter with cut off [4, 12] Hz and [0.5, 8] Hz, respectively. The filtered PPG and BCG signals are segmented into non-overlapped windows with a window length of 12 seconds. We designed a decision tree to use the blood pulse in the PPG signal to determine the fiducial points. Specifically, the I-wave represents the major downward deflection of the BCG associated with the opening of the aortic valve, while the H-wave and J-wave are the preceding and succeeding prominent peaks, respectively. An example illustrating these identified waves along with their timing relative to the ground truth B-point is shown in Fig. 6. The algorithm illustrating I-wave detection is illustrated in Fig. 7.

D. Multi-axis BCG fusion from Earbuds

Although the orientation of the IMU chip set was designed to align the accelerometer's Y-axis with the user's head-foot direction, we found that the X-axis carries identical or complementary information for many subjects. To enhance SNR, we developed an algorithm for BCG fusion using both

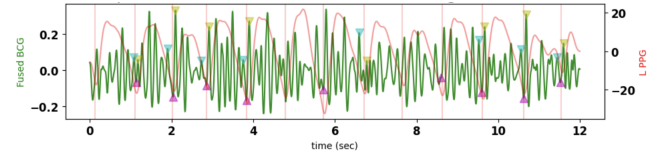


Fig. 6. The identified H-wave (as indicated by the triangle in the cyan color), I-wave (as indicated by the triangle in the magenta color), and J-wave (as indicated by the triangle in the yellow color) in a representative BCG signal (the signal as shown in the green color). The overlapped red signal is the PPG signal measured from the same earbuds. The vertical bars represent ground truth opening of aortic valve as identified jointly using ECG and ICG signals.

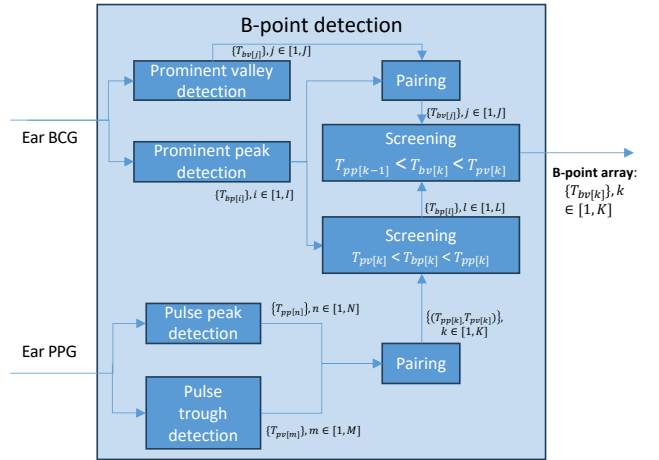


Fig. 7. The flowchart for determining I-wave by jointly using the synchronized PPG and BCG signals.

the X-axis and Y-axis. First, individual filtered BCG signals are normalized in the range between 0 and 1. Second, the normalized signals are multiplexed to form the fused BCG signal. The effect of this BCG fusion is illustrated in Fig. 8, in which the originally noisy BCG signals in both the x and y axes are fused together. The BCG fusion process improved the SNR so that the B-point can be identified using the algorithm illustrated in Fig. 7.

E. Comparison Between Estimated and Ground Truth B-points

By considering the H-wave, I-wave, or J-wave as the proxy of B-point, respectively, we estimated the timing of B-points for all valid heartbeats, all stages, and all participants. For each of the candidate proxy, we used the BCG Fusion, the BCG X-axis, and BCG Y-axis, separately. We then calculated the time difference between the estimated B-point and the ground truth B-point for each beat. We consider that a good estimation of B-points should have no inherent bias towards positive or negative values, when compared to the ground truth. The distributions of Δt are illustrated as histograms in Fig. 9. It shows that Δt is the most symmetric when we use the I-wave as the proxy, as illustrated in the second row. The histograms associated with the H-wave (shown in the 1st row) and the J-wave (shown in the 3rd row) are biased towards either positive or negative, showing systematic error

TABLE II
AVERAGED MAE BETWEEN ESTIMATED AND GROUND TRUTH B-POINTS ACROSS PARTICIPANTS.

Candidate Proxy of B-points	Δt using BCG Fusion (msec)				Δt using BCG X-axis (msec)				Δt using BCG Y-axis (msec)			
	Average	STD	Skewness	MAE	Average	STD	Skewness	MAE	Average	STD	Skewness	MAE
H-wave	13.7	43.99	-0.502	94.06	6.99	43.6	-0.274	91.04	9.37	42.03	-0.428	88.31
I-wave	-1.05	17.58	0.179	38.14	1.64	18.98	0.002	40.29	-2.79	18.2	0.373	40.88
J-wave	-14.05	23.59	1.017	56.16	-13.28	46.23	0.453	96.68	-13.1	43.35	0.672	94.06

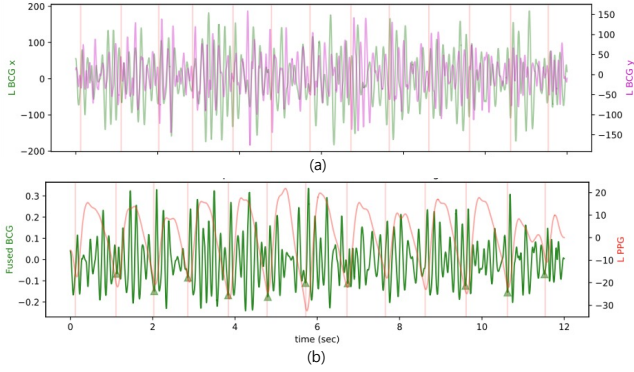


Fig. 8. (a) Exemplary BCG signals from BCG X-axis (shown in magenta color) and Y-axis (shown in green color) from a plateau session. The signals from either axis are too noisy to determine the B-point. (b) After BCG fusion, the B-points are visually identifiable (as indicated by triangles in green color) and can be picked up using the algorithm illustrated in Fig. 7.

when used as a proxy.

We employed the two one-sided test (TOST) to verify whether the mean of Δt is effectively zero [16]. The null hypothesis assumes that the mean is nonzero (biased to either positive or negative), while the alternative hypothesis posits that the mean falls within a margin of $\pm 2 \text{ msec}$. We conducted two one-sided t-tests: one testing if the mean is significantly greater than -2 msec and the other if it is significantly less than 2 msec . The p -values for both tests are shown in the legend of each histogram. The result shows that only when we use the I-wave as the proxy and derive B-point using the BCG fusion, both the tests reject their respective null hypotheses ($\alpha = 0.05$), showing that the Δt is not biased.

To quantify the accuracy in the detection of B-points, we calculated the average and the standard deviation (STD) of Δt when using different combinations of B-point proxy and BCG axis. To calculate the averaged mean absolute error (MAE), we first calculate the MAE within each stage, then calculate the averaged and standard deviation of the MAE across stages and participants. The results are reported in Table II. It is shown that when using the I-wave as the proxy and estimated on the basis of the BCG fusion method, we obtain the least MAE across all combinations. Specifically, when using the I-wave derived from the BCG fusion as a proxy of the B-points, we reduced MAE by 5.33% and 6.70% compared to the I-wave derived only from the X-axis BCG and the Y-axis BCG, respectively. Meanwhile, using the I-wave as the proxy B-point reduces MAE by 59.45%

and 32.09% compared to using H-wave and the J-wave as the proxy, respectively.

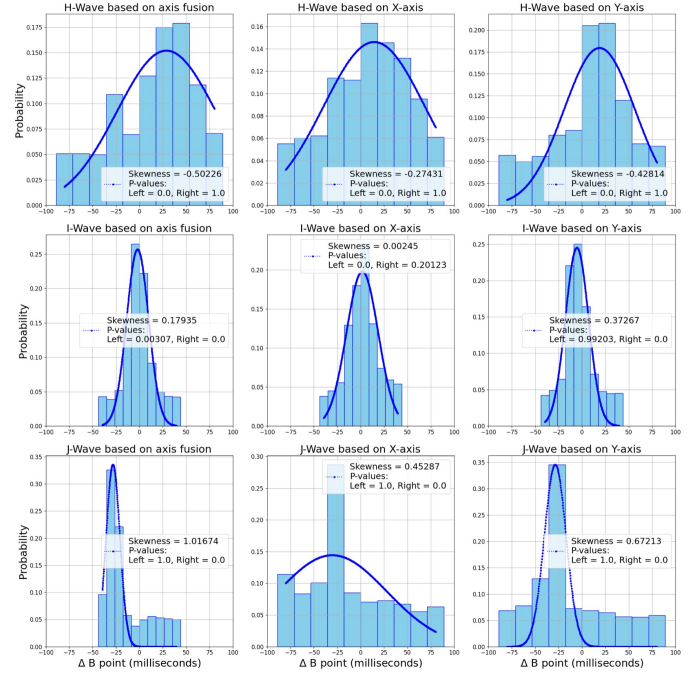


Fig. 9. The histograms showing Δt between estimated and ground truth B-points. The blue curve is the fitted Gaussian function. The first row: Δt based on H-waves. The second row: Δt based on I-waves. The third row: Δt based on J-waves. The first column: Δt based on fused BCG. The second column: Δt based on X-axis of BCG. The third column: Δt based on Y-axis of BCG.

IV. CONCLUSION

This study introduces an innovative earbud-based system designed for continuous and noninvasive BP monitoring using precise PTT measurements. The system integrates a low-noise IMU with a synchronized PPG sensor, allowing direct capture of the mechanical event of aortic valve opening in the concha area of the ear location. The key to this novel approach is the precise detection of the aortic valve opening, which is achieved through the analysis of the BCG waveform. The I-wave of the BCG signal was rigorously validated against the ground truth opening time of the aortic valve derived from synchronized ECG and ICG, proving to be the most reliable proximal timing marker for PTT-based estimation of BP. The use of a multiaxis BCG fusion algorithm further enhances the SNR and the accuracy of B-point detection. The system's synchronization accuracy of less than 1.1 milliseconds between the IMU and PPG

sensors is crucial for minimizing BP estimation errors. However, the generalizability of the findings could be limited by the specific characteristics of the study population and the posture of the user. Additionally, slight variations in earbud positioning might influence the consistency of BCG measurements. Future studies should evaluate the robustness of the system across diverse populations and various user postures in real-world environments.

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